

Student Name: \_\_\_\_\_

Teacher: \_\_\_\_\_

2024

Hurlstone Agricultural High  
School  
Year 12 Assessment Task 2

# Mathematics Extension 2

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**Examiners**

- **Mr D Potaczala**
- **Mr G Rawson**
- **Mr G Huxley**

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**General  
Instructions**

- Reading time – 3 minutes
- Working time – 40 minutes
- Write using black or blue pen
- NESA-approved calculators may be used
- A Reference sheet is provided for your use
- An A4 handwritten summary sheet may be used
- In Questions 7 to 9, show relevant mathematical reasoning and/or calculations

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**Total marks:  
30****Section I – 6 marks (page 2–3)**

- Attempt Questions 1–6
- Allow about 8 minutes for this section

**Section II – 24 marks (pages 4–6)**

- Attempt Questions 7–9
- Allow about 32 minutes for this section
- For each question use a separate answer booklet

## Section I

**6 marks**

**Attempt Questions 1 to 6**

**Allow about 8 minutes for this section**

Use the multiple-choice answer sheet for Questions 1 to 6

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1. Consider the statement:  
For any function  $f(x)$ ,  $f(x)$  has a stationary point of inflection at  $x = c \Rightarrow f''(c) = 0$ .
- Which of the following is correct?
- A. The contrapositive statement is false and the converse statement is false.
  - B. The contrapositive statement is false and the converse statement is true.
  - C. The contrapositive statement is true and the converse statement is false.
  - D. The contrapositive statement is true and the converse statement is true.
2. Consider the statement:  
“ $\forall x \in \mathbb{N}$ ,  $x$  is even  $\Rightarrow x^2(x^2 + 4)$  is divisible by 16.”
- Which of the following is the negation of the statement?
- A.  $\exists x \in \mathbb{N}$ ,  $x$  is not even and  $x^2(x^2 + 4)$  is not divisible by 16.
  - B.  $\exists x \in \mathbb{N}$ ,  $x$  is even and  $x^2(x^2 + 4)$  is not divisible by 16.
  - C.  $\forall x \in \mathbb{N}$ ,  $x$  is even and  $x^2(x^2 + 4)$  is not divisible by 16.
  - D.  $\forall x \in \mathbb{N}$ ,  $x$  is even or  $x^2(x^2 + 4)$  is not divisible by 16.

3. Consider the proposition:

‘If  $2^n - 1$  is not prime, then  $n$  is not prime’

Given that each of the following statements is true, which statement disproves the proposition?

- A.  $2^5 - 1$  is prime
- B.  $2^6 - 1$  is divisible by 9
- C.  $2^7 - 1$  is prime
- D.  $2^{11} - 1$  is divisible by 23

4. Which of the following is the principal fifth root of  $-2$ ?

- A.  $e^{i\pi}$
- B.  $e^{\frac{i\pi}{5}}$
- C.  $\sqrt[5]{2}e^{i\pi}$
- D.  $\sqrt[5]{2}e^{\frac{i\pi}{5}}$

5. What are the square roots of  $5 - 12i$  ?

- A.  $\pm(3 - 2i)$
- B.  $\pm(2 + 3i)$
- C.  $\pm(2 - 3i)$
- D.  $\pm(3 + 2i)$

6. If  $z$  is a complex number such that  $\frac{z-1}{z+1}$  is purely imaginary, then what is  $|z|$  equal to?

- A.  $\frac{1}{2}$
- B.  $\frac{2}{3}$
- C. 1
- D. 2

~ END OF SECTION I ~

## Section II

**18 marks**

**Attempt Questions 7 to 9**

**Allow about 32 minutes for this section**

*Answer each question in a new answer booklet.*

All necessary working should be shown in every question.

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### Question 7 (8 marks)

**Marks**

- (a) If  $x \leq 1$  and  $y \geq 1$ , prove that  $x + y \geq 1 + xy$ . **2**
- (b) Given that  $a$ ,  $b$  and  $c$  are positive real numbers, prove that **3**
- $$(a + b + c)^2 \geq 3(ab + ac + bc).$$
- (c) Prove, using contradiction, that  $\sqrt{4n - 2}$  is irrational for all positive integers,  $n$ . **3**

*~ End of question 7 ~*

**Question 8 (8 marks)****Marks**

- (a) (i) Sketch the set of points on an Argand Diagram, for which:

**1**

$$\arg(z) = \frac{\pi}{3}$$

- (ii) Show on your sketch from (i) why the distance given by  $|z - 2i|$  is greater than or equal to 1 unit.

**1**

- (iii) Find the value of  $z$  for which  $|z - 2i| = 1$ .  
Give polar coordinates for your answer.

**1**

- (b) Let  $P(z) = 2z^3 - 13z^2 + 26z - 10$ .

- (i) Show that  $(z - 3 + i)$  is a factor of  $P(z)$ .

**2**

- (ii) Hence or otherwise, solve  $P(z) = 0$ .

**2**

- (iii) Write  $P(z)$  as a product of factors with real integer coefficients.

**1**

*~ End of question 8 ~*

**Question 9 (8 marks)****Marks**

- (a) Prove by induction that, for all integers  $n \geq 1$ , **3**

$$(\cos \theta - i \sin \theta)^n = \cos(n\theta) - i \sin(n\theta)$$

- (b) Suppose that  $z = \frac{1}{2}(\cos \theta + i \sin \theta)$  where  $\theta$  is real.

- (i) Show that the imaginary part of the geometric series **3**

$$1 + z + z^2 + z^3 + \dots = \frac{1}{1 - z}$$

is  $\frac{2 \sin \theta}{5 - 4 \cos \theta}$ .

- (ii) Find an expression for **2**

$$1 + \frac{1}{2} \cos \theta + \frac{1}{2^2} \cos 2\theta + \frac{1}{2^3} \cos 3\theta + \dots$$

in terms of  $\cos \theta$ .

**~ END OF SECTION II ~**

**~ END OF PAPER ~**

Mathematics Advanced  
Mathematics Extension 1  
Mathematics Extension 2

REFERENCE SHEET

**Measurement**

**Length**

$$l = \frac{\theta}{360} \times 2\pi r$$

**Area**

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

**Surface area**

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

**Volume**

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

**Functions**

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For  $ax^3 + bx^2 + cx + d = 0$ :

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\text{and } \alpha\beta\gamma = -\frac{d}{a}$$

**Relations**

$$(x - h)^2 + (y - k)^2 = r^2$$

**Financial Mathematics**

$$A = P(1 + r)^n$$

**Sequences and series**

$$T_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{a(r^n - 1)}{r - 1}, r \neq 1$$

$$S = \frac{a}{1 - r}, |r| < 1$$

**Logarithmic and Exponential Functions**

$$\log_a a^x = x = a^{\log_a x}$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$a^x = e^{x \ln a}$$

## Trigonometric Functions

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab \sin C$$

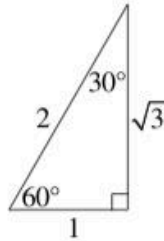
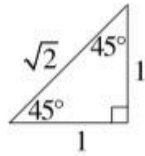
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$l = r\theta$$

$$A = \frac{1}{2}r^2\theta$$



## Trigonometric identities

$$\sec A = \frac{1}{\cos A}, \quad \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \quad \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \quad \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

## Compound angles

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\text{If } t = \tan \frac{A}{2} \text{ then } \sin A = \frac{2t}{1+t^2}$$

$$\cos A = \frac{1-t^2}{1+t^2}$$

$$\tan A = \frac{2t}{1-t^2}$$

$$\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$$

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

$$\sin A \cos B = \frac{1}{2} [\sin(A + B) + \sin(A - B)]$$

$$\cos A \sin B = \frac{1}{2} [\sin(A + B) - \sin(A - B)]$$

$$\sin^2 nx = \frac{1}{2} (1 - \cos 2nx)$$

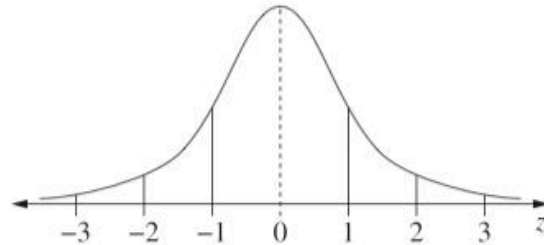
$$\cos^2 nx = \frac{1}{2} (1 + \cos 2nx)$$

## Statistical Analysis

$$z = \frac{x - \mu}{\sigma}$$

An outlier is a score  
less than  $Q_1 - 1.5 \times IQR$   
or  
more than  $Q_3 + 1.5 \times IQR$

## Normal distribution



- approximately 68% of scores have z-scores between -1 and 1
- approximately 95% of scores have z-scores between -2 and 2
- approximately 99.7% of scores have z-scores between -3 and 3

$$E(X) = \mu$$

$$\operatorname{Var}(X) = E[(X - \mu)^2] = E(X^2) - \mu^2$$

## Probability

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$$

## Continuous random variables

$$P(X \leq x) = \int_a^x f(x) dx$$

$$P(a < X < b) = \int_a^b f(x) dx$$

## Binomial distribution

$$P(X = r) = {}^nC_r p^r (1-p)^{n-r}$$

$$X \sim \operatorname{Bin}(n, p)$$

$$\Rightarrow P(X = x)$$

$$= {}^nC_x p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n$$

$$E(X) = np$$

$$\operatorname{Var}(X) = np(1-p)$$

## Differential Calculus

### Function

### Derivative

$$y = f(x)^n$$

$$\frac{dy}{dx} = n f'(x) [f(x)]^{n-1}$$

$$y = uv$$

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$y = g(u) \text{ where } u = f(x)$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$y = \frac{u}{v}$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$y = \sin f(x)$$

$$\frac{dy}{dx} = f'(x) \cos f(x)$$

$$y = \cos f(x)$$

$$\frac{dy}{dx} = -f'(x) \sin f(x)$$

$$y = \tan f(x)$$

$$\frac{dy}{dx} = f'(x) \sec^2 f(x)$$

$$y = e^{f(x)}$$

$$\frac{dy}{dx} = f'(x) e^{f(x)}$$

$$y = \ln f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

$$y = a^{f(x)}$$

$$\frac{dy}{dx} = (\ln a) f'(x) a^{f(x)}$$

$$y = \log_a f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{(\ln a) f(x)}$$

$$y = \sin^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \cos^{-1} f(x)$$

$$\frac{dy}{dx} = -\frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \tan^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{1 + [f(x)]^2}$$

## Integral Calculus

$$\int f'(x) [f(x)]^n dx = \frac{1}{n+1} [f(x)]^{n+1} + c$$

where  $n \neq -1$

$$\int f'(x) \sin f(x) dx = -\cos f(x) + c$$

$$\int f'(x) \cos f(x) dx = \sin f(x) + c$$

$$\int f'(x) \sec^2 f(x) dx = \tan f(x) + c$$

$$\int f'(x) e^{f(x)} dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c$$

$$\int f'(x) a^{f(x)} dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \sin^{-1} \frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2 + [f(x)]^2} dx = \frac{1}{a} \tan^{-1} \frac{f(x)}{a} + c$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int_a^b f(x) dx$$

$$\approx \frac{b-a}{2n} \left\{ f(a) + f(b) + 2[f(x_1) + \dots + f(x_{n-1})] \right\}$$

where  $a = x_0$  and  $b = x_n$

## Combinatorics

$${}^nP_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}^nC_r = \frac{n!}{r!(n-r)!}$$

$$(x+a)^n = x^n + \binom{n}{1}x^{n-1}a + \cdots + \binom{n}{r}x^{n-r}a^r + \cdots + a^n$$

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## Vectors

$$|\underline{u}| = |x\underline{i} + y\underline{j}| = \sqrt{x^2 + y^2}$$

$$\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos \theta = x_1x_2 + y_1y_2,$$

$$\text{where } \underline{u} = x_1\underline{i} + y_1\underline{j}$$

$$\text{and } \underline{v} = x_2\underline{i} + y_2\underline{j}$$

$$\underline{r} = \underline{a} + \lambda \underline{b}$$

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## Complex Numbers

$$\begin{aligned} z &= a + ib = r(\cos \theta + i \sin \theta) \\ &= re^{i\theta} \end{aligned}$$

$$\begin{aligned} [r(\cos \theta + i \sin \theta)]^n &= r^n(\cos n\theta + i \sin n\theta) \\ &= r^n e^{in\theta} \end{aligned}$$

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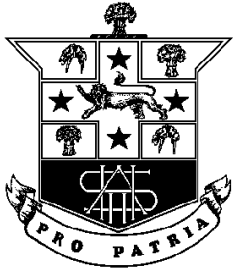
## Mechanics

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left( \frac{1}{2} v^2 \right)$$

$$x = a \cos(nt + \alpha) + c$$

$$x = a \sin(nt + \alpha) + c$$

$$\ddot{x} = -n^2(x - c)$$



Student Name: \_\_\_\_\_

Teacher: \_\_\_\_\_

2024

Hurlstone Agricultural High School  
Year 12 Mathematics Extension 2  
Assessment Task 2

Section I – Multiple Choice Answer Sheet

Allow about 10 minutes for this section

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample:  $2 + 4 =$  (A) 2 (B) 6 (C) 8 (D) 9  
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

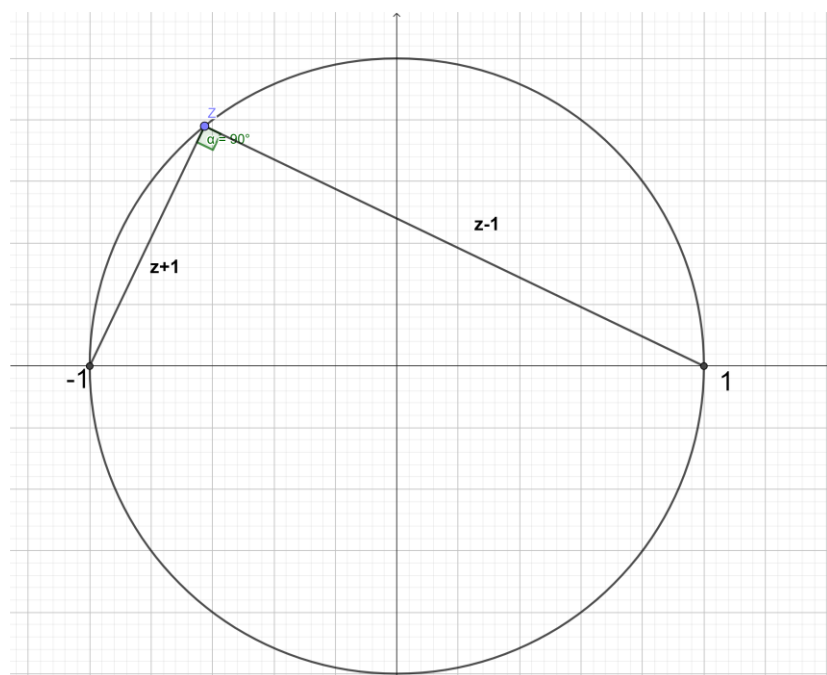
A ☒ B ☒ C ☐ D ☐  
correct

Start Here

1. A ☐ B ☐ C ☐ D ☐
2. A ☐ B ☐ C ☐ D ☐
3. A ☐ B ☐ C ☐ D ☐
4. A ☐ B ☐ C ☐ D ☐
5. A ☐ B ☐ C ☐ D ☐
6. A ☐ B ☐ C ☐ D ☐

| 2024 Yr12 HSC Assessment Task 2 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                    |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| Multiple Choice                 | Solutions and Marking Guidelines                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                    |
|                                 | Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Marking Guidelines |
| 1)                              | <p>Contrapositive: <math>f''(x) \neq 0 \Rightarrow f(x)</math> does not have a stationary point of inflection at <math>x = c</math>. TRUE</p> <p>Converse: <math>f''(x) \neq 0 \Rightarrow f(x)</math> has a stationary point of inflection at <math>x = c</math>. FALSE</p> <p>Correct answer is C</p>                                                                                                                                                                                                                                                              |                    |
| 2)                              | <p>The negation is B</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                    |
| 3)                              | <p>Option D</p> <p>Since 11 is prime and <math>2^{11} - 1</math> is not prime as it is divisible by 23</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                    |
| 4)                              | <p>Finding the fifth root of <math>-2</math> is to solve the equation <math>z^5 = -2</math>.</p> <p>Using <math>e^{i\pi} = -1</math> we have <math>z^5 = 2e^{i\pi}</math></p> <p>Hence, it is clear, with not further working required, that option D is the answer.</p>                                                                                                                                                                                                                                                                                             |                    |
| 5)                              | $(x + iy)^2 = 5 - 12i$ <p>The 4 options all give 2 and 3 as components of either real or imaginary parts, so we need to concentrate on the required relationship between the components.</p> <p><math>2xy = -12</math> and <math>x^2 - y^2 = 5</math> give <math>x = \pm 3; y = \mp 2</math></p> <p>Answer A</p> <p><i>Note: As an extra check: With what we know about doubling the argument when we square a number: since <math>5-12i</math> is in the 4<sup>th</sup> quadrant, then one of its square roots must also be in the 4<sup>th</sup> quadrant.</i></p> |                    |

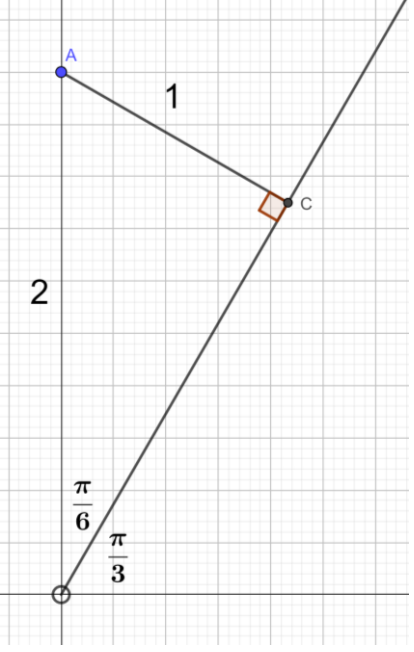
6)



The question implies that the vectors of  $z-1$  and  $z+1$  are perpendicular. Hence the complex number  $z$  must lie on the circle with diameter joining  $-1$  and  $1$ . Hence  $|z| = 1$ . Answer C.

| Year 12                                                                                                                  | Mathematics Extension 2                                                                                                                                                                                                                                                                                                                 | Ass Task 2 2024 HSC                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Question 7                                                                                                               | Solutions and Marking Guidelines                                                                                                                                                                                                                                                                                                        |                                                                                                                                                |
| <b>Outcomes Addressed in this Question</b>                                                                               |                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                |
| <b>MEX 12-2</b> chooses appropriate strategies to construct arguments and proofs in both practical and abstract settings |                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                |
| Question/<br>Outcome                                                                                                     | Solutions                                                                                                                                                                                                                                                                                                                               | Marking Guidelines                                                                                                                             |
| (a)                                                                                                                      | $x \leq 1 \rightarrow x - 1 \leq 0$ $y \geq 1 \rightarrow y - 1 \geq 0$ $(x - 1)(y - 1) \leq 0$ $xy - x - y + 1 \leq 0$ $xy + 1 \leq x + y$ $\therefore x + y \geq 1 + xy$                                                                                                                                                              | <p><u>2 marks</u>: correct solution</p> <p><u>1 mark</u>: substantially correct solution</p>                                                   |
| (b)                                                                                                                      | $(a - b)^2 \geq 0 \Rightarrow a^2 + b^2 \geq 2ab$ $(b - c)^2 \geq 0 \Rightarrow b^2 + c^2 \geq 2bc$ $(a - c)^2 \geq 0 \Rightarrow a^2 + c^2 \geq 2ac$ $2a^2 + 2b^2 + 2c^2 \geq 2ab + 2bc + 2ac$ $a^2 + b^2 + c^2 \geq ab + bc + ac$ $(a + b + c)^2 - 2ab - 2bc - 2ac \geq ab + bc + ac$ $\therefore (a + b + c)^2 \geq 3ab + 3bc + 3ac$ | <p><u>3 marks</u>: correct solution</p> <p><u>2 marks</u>: substantially correct solution</p> <p><u>1 mark</u>: partially correct solution</p> |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (c) | <p>Assume, by way of contradiction, that <math>\sqrt{4n-2}</math> is rational<br/> ie <math>\sqrt{4n-2} = \frac{p}{q}</math>, where <math>p</math> and <math>q</math> have highest common factor 1,<br/> and <math>q \neq 0</math>.</p> $\therefore 4n - 2 = \frac{p^2}{q^2}$ $2(2n - 1) = \frac{p^2}{q^2}$ <p>Since LHS is an even integer, <math>\frac{p^2}{q^2}</math> must also be an even integer.</p> <p>Since HCF of <math>p</math> and <math>q</math> is 1, then HCF of <math>p^2</math> and <math>q^2</math> is also 1.</p> <p>Therefore <math>q^2 = 1</math> such that <math>\frac{p^2}{q^2}</math> is an even integer.</p> $\therefore 4n - 2 = \frac{p^2}{q^2} = p^2$ <p>Hence, <math>p^2</math> is even, and therefore <math>p</math> is also even.<br/> ie <math>p = 2k</math> where <math>k</math> is an integer.</p> $\therefore 4n - 2 = (2k)^2$ $4n - 2 = 4k^2$ $n - \frac{1}{2} = k^2$ $k = \sqrt{n - \frac{1}{2}}$ <p><math>\therefore k</math> is <i>not</i> an integer.</p> <p>This is a contradiction (as <math>p = 2k</math> where <math>k</math> is an integer)</p> <p>Hence the initial premise is false, and <math>\sqrt{4n-2}</math> is <b>not</b> rational</p> | <p><u>3 marks</u>: correct solution</p> <p><u>2 marks</u>: substantially correct solution (identifies <b><u>and uses</u></b> the factors of <math>p^2</math> and <math>q^2</math> and the idea of even/odd integers to establish the base premise leading up to the contradiction statement)</p> <p><u>1 mark</u>: partially correct solution (establishes the correct initial premise for a contradiction based proof, <b><u>including</u></b> the condition on <math>p</math> and <math>q</math>)</p> <p>Note: a common statement in solutions was along the lines of</p> $2p^2 = b^2(2n - 1)$ <p><math>\therefore b^2</math> is even</p> <p>There needs to be an extra step of explanation as to <i>why</i> this is the case,<br/> eg noting that <math>2n - 1</math> is odd, and odd x even = even</p> |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

| Higher School Certificate                                                                                                                                                 | Mathematics Extension 2                                                                                                                                                                                                                                                                                                                                                                                                          | Task 2 2024 HSC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Solutions and Marking Guidelines                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Question 8                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Outcomes Addressed in this Question                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| MEX12-4 uses the relationship between algebraic and geometric representations of complex numbers and complex number techniques to prove results, model and solve problems |                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Question                                                                                                                                                                  | Solutions                                                                                                                                                                                                                                                                                                                                                                                                                        | Marking Guidelines                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| (a)(i)(ii)                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                | <p><b>(a)(i) 1 mark:</b> Correct ray from origin with angle labelled and open circle at the origin.</p> <p><i>Note: The majority of responses did not include the open circle at the origin. This is a small but crucial detail when illustrating the boundary of an argument.</i></p> <p><b>(ii) 1 mark:</b> Perpendicular segment indicated, creating “exact value” right triangle with hypotenuse 2 units and angle <math>\pi/6</math>, thus the perpendicular distance will equal 1 unit.</p> <p><b>Aw1:</b> correct <u>perpendicular</u> segment indicated.</p> |
| (iii)                                                                                                                                                                     | <p><math>\arg(z) = \frac{\pi}{3}</math> , <math>\text{mod}(z) = \sqrt{3}</math> due to the exact values in the right triangle.</p> $\therefore z = \sqrt{3} \left( \cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$ <p><i>Note: Several responses to (iii) featured a circle that had no relation to the answers to (i) and (ii). Exam method requires understanding of the relationship within part (a) in this case.</i></p> | <p><b>(iii)</b> Correct answer. <i>Note: “Polar coordinates” are the modulus and argument.</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

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| <p><b>(b)</b></p> <p><b>(i)</b></p> <p><b>(ii)</b></p> <p><b>(iii)</b></p> | $P(z) = 2z^3 - 13z^2 + 26z - 10$ $P(3 - i) = 2(3 - i)^3 - 13(3 - i)^2 + 26(3 - i) - 10$ $= 2(18 - 26i) - 13(8 - 6i) + 26(3 - i) - 10$ $= 36 - 52i - 104 + 78i + 78 - 26i - 10$ $= 0 + 0i$ <p>Since <math>P(3 - i) = 0</math>, <math>(z - 3 + i)</math> is a factor of <math>P(z)</math>.</p> <p>All coefficients are real, so the complex roots occur in complex conjugate pairs. Use either the sum of the roots or the product of the roots to find the third root of the polynomial equation <math>P(z) = 0</math>.</p> <p>Either: <math>3 + i + 3 - i + \alpha = \frac{13}{2} \rightarrow \alpha = \frac{1}{2}</math></p> <p>Or: <math>(3 + i)(3 - i)\alpha = \frac{10}{2} \rightarrow 10\alpha = 5 \rightarrow \alpha = \frac{1}{2}</math></p> <p><math>\therefore P(z) = 0</math> gives <math>z = 3 \pm i, \frac{1}{2}</math></p> <p><i>Note: Responses need to answer the question by clearly identifying the 3 values that solve <math>P(z)=0</math>. Some sloppy working out was given the benefit of the doubt in this case but that may not always occur. Remember to give the best answer you can to increase the likelihood of marks being awarded.</i></p> <p>The response needs to be a product of factors within the real number field, and all coefficients are to be integral.</p> <p><math>\alpha = \frac{1}{2}</math> and a leading coefficient of 2 gives one factor equal to <math>(2z - 1)</math></p> <p>Hence <math>P(z)</math> factorised over the real numbers will be:</p> $(2z - 1)((z - 3) - i)((z - 3) + i)$ $= (2z - 1)(z^2 - 6z + 9 + 1)$ $= (2z - 1)(z^2 - 6z + 10)$ | <p><b>(b)(i) 2 marks:</b> Correct solution with all relevant working shown.</p> <p><b>1 mark:</b> Significant relevant progress and understanding communicated.</p> <p><b>(ii) 2 marks:</b> Acknowledgement of conjugate theorem and of sum/product of roots; Correct calculation of third root.</p> <p><b>1 mark:</b> One component from above completed.</p> <p><b>(iii) Correct solution</b><br/>CFPA (ii)</p> |
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| 2023 Yr12 HSC Assessment Task 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                       |
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| Question 9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Solutions and Marking Guidelines                                                                                                                                                                                                                                                                                                                      |
| <b>Outcomes Addressed in this Question</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                       |
| MEX12-4<br>uses the relationship between algebraic and geometric representations of complex numbers and complex number techniques to prove results, model and solve problems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                       |
| Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marking Guidelines                                                                                                                                                                                                                                                                                                                                    |
| <p>a)</p> <p>Show true for base case <math>n = 1</math></p> $LHS = (\cos \theta - i \sin \theta)^1$ $RHS = \cos(1 \times \theta) - i \sin(1 \times \theta)$ $= LHS$ <p>Hence, true for <math>n = 1</math>.</p> <p>Assume the statement is true for <math>n = k</math> where <math>k</math> is some positive integer. That is:</p> $(\cos \theta - i \sin \theta)^k = \cos(k\theta) - i \sin(k\theta)$ <p>Prove the statement true for <math>n = k + 1</math>. That is:</p> $(\cos \theta - i \sin \theta)^{k+1} = \cos((k + 1)\theta) - i \sin((k + 1)\theta)$ $LHS = (\cos \theta - i \sin \theta) (\cos \theta - i \sin \theta)^k$ $= (\cos \theta - i \sin \theta) (\cos(k\theta) - i \sin(k\theta)) \quad (\text{from assumption})$ $= \cos(k\theta) \cos(\theta) - \sin(k\theta) \sin(\theta) - i(\sin(k\theta) \cos(\theta) + \cos(k\theta) \sin(\theta))$ $= \cos(k\theta + \theta) - i \sin(k\theta + \theta) \quad (\text{compound angle identity})$ $= \cos((k + 1)\theta) - i \sin((k + 1)\theta)$ <p><math>\therefore</math> if true for <math>n = k</math>, it is also true for <math>n = k + 1</math>.</p> <p><math>\therefore</math> statement true by Mathematical Induction, for all positive integers.</p> <p>b) i)</p> | <p><b>3 marks</b><br/>Correct solution</p> <ul style="list-style-type: none"> <li>•Includes test for <math>n=1</math></li> <li>•Uses induction assumption</li> <li>•Applies correct trigonometric identity</li> </ul> <p><b>2 marks</b><br/>Establishes inductive step and uses assumption</p> <p><b>1 mark</b><br/>Correctly tests the base case</p> |

$$\begin{aligned}
\frac{1}{1-z} &= \frac{1}{1 - \frac{1}{2}(\cos \theta + i \sin \theta)} \\
&= \frac{2}{2 - \cos \theta - i \sin \theta} \times \frac{2 - \cos \theta + i \sin \theta}{2 - \cos \theta + i \sin \theta} \\
&= \frac{2(2 - \cos \theta + i \sin \theta)}{(2 - \cos \theta)^2 + \sin^2 \theta} \\
&= \frac{4 - 2 \cos \theta + 2i \sin \theta}{4 - 4 \cos \theta + \cos^2 \theta + \sin^2 \theta} \\
&= \frac{4 - 2 \cos \theta + 2i \sin \theta}{5 - 4 \cos \theta}
\end{aligned}$$

Taking the imaginary part, we have:  $\frac{2i \sin \theta}{5 - 4 \cos \theta}$

Note: you can only take the imaginary part of a fraction involving complex numbers if the denominator is *realised*.

ii)

It can be seen that

$$1 + \frac{1}{2} \cos \theta + \frac{1}{2^2} \cos \theta + \frac{1}{2^3} \cos \theta + \dots$$

is the real part of the geometric series in part (i).

Hence, an expression for it can be

$$\frac{4 - 2 \cos \theta}{5 - 4 \cos \theta}$$

1 mark explicitly links expression to real part of i, 1 mark find correct expression.

**3 marks**  
Correct solution

**2 marks**  
Correctly realises the denominator

**1 mark**  
Uses RHS part of geometric series

**2 marks**  
Correct solution including reference to part (i)

**1 mark**  
Makes substantial progress towards correct solution